Lalit Negi, M.Tech

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BIOGRAPHY

I am a researcher with a background in Geographic Information Systems (GIS), Augmented/Virtual Reality (AR/VR), simulation and applied cryptography, now redirecting my work towards LGBTQ+ communities. I am interested in how LGBTQ+ people experience safety, belonging and inclusion, and in how spatial, immersive and computational technologies can be designed with and for them. My earlier work in applied cryptography and information security provides a foundation for privacy, security and the responsible handling of sensitive LGBTQ+ data across all my research areas.

RESEARCH INTERESTS

- **Queer GIS and Critical Geographies:** Spatial analysis, web and 3D GIS, simulation and GeoAI to map disparities in rights and protections, spatial patterns of discrimination, and access to LGBTQ+-supportive services and spaces, and to examine how places influence safety, belonging, participation and quality of life.
- **Social XR and HCI:** Shared virtual, augmented and mixed reality spaces for identity expression, peer support and community, designed through participatory and community-centred HCI, with attention to presence, safety, accessibility, harassment mitigation and privacy.
- **Digital Humanities and Critical Computing:** The history, memory and lived experiences of LGBTQ+ communities studied through digital archives, heritage and community-led data, together with a critical look at bias, representation and privacy in datasets, algorithms and large language models.

EXPERIENCE

Central Research Laboratory, Bharat Electronics Limited *India, Ghaziabad*
Member Research Staff *2022 – Present*

GIS Tech Lab *Jan 2024- Present*

- Currently working on web based **CIDSS** Command Information and Decision Support System for the Indian Army using the **Luciad RIA sdk package**.
- Leading the team working on the development of **VR Based Digital Sandtable** prototype using the **ArcGIS sdk** in Unity
- Designed a virtual Tactical Theatre Command system using **ArcGIS**, visualizing real-world deployment of defence resources, operational assets, and live tracks to support situational awareness and command-level planning.
- Successfully participated in the demonstration of the **Tactical Theatre Command System** to the **Honourable Defence Minister of India**, showcasing real-time visualization, operational planning, and decision-support capabilities.

Decision Support System Lab *July 2022 - Dec 2023*

- Contributed to the development of the **first indigenously developed universal train driving simulator** through simulation algorithm design and secure communication protocol development.
- Developed simulation modules such as **Simulation Engine**, **Training and Evaluation Assessment Module**, and **Malfunction Handler** in **C++** using **Qt** to model metro train logic, control, and operational behavior.
- Prepared technical documentation including **Software Requirements Specification (SRS)** and **Software Requirement Description (SRD)** to support project delivery and audits.
- Developed a **VR-based training simulator** for radar assembly and disassembly, enabling step-by-step procedural training in an immersive and safe virtual environment.
- Developed a **SR-based display system** to create a **Digital twin** of radar systems for predictive maintenance, condition monitoring, and fault analysis.

PERSONAL PROJECTS

- **Rainbow Connect** *(HCI, Inclusive Technology)*
  - Theme: A web-based communication platform designed to support LGBTQ+ communities in offices and institutions. The platform enables users to connect through text-based communication and video calling, with support for both Hindi and English. Designed as an inclusive digital space to encourage communication, connection, and peer interaction within institutional environments.
- **Indradhanush Spaces** *(Queer GIS, Inclusive Technology)*
  - Theme: A web-based GIS application for discovering and registering LGBTQ+-inclusive spaces. Space owners can register their inclusive spaces on the platform, while users can explore nearby spaces through an interactive map. The application uses location-based visualization to display spaces around the user's current location and identify places within a selected geographic range.
- **OpenQueer Data** *(Critical Computing, Community Data)*
  - Theme: A web-based dataset management application for Queer datasets designed to create new datasets, populate and update existing datasets, and manage structured data efficiently. It also provides options to review, edit, and download datasets in a convenient format for further analysis and processing.
- **Geographic Information System-Based Real-Time Fire Detection and Anti-Fire Simulation System**
  - Theme: A GIS-based system for real-time fire detection, spatial visualization, and fire-spread simulation. Integrates geospatial data, real-time information, and simulation models to support fire monitoring, prediction, and emergency response planning.
- **Interactive Radar Fault Management Training System with GIS-Based Visualization and Corrective Action Guidance**
  - Theme: An interactive training system that integrates GIS-based visualization with radar fault scenarios and corrective-action guidance. Designed to provide an immersive and interactive environment for fault identification, decision-making, and technical training.
- **Immersive Virtual Reality Visualization of Machine Learning-Based Weather Classification for Climate Scenario Analysis**
  - Theme: An immersive virtual reality environment for visualizing machine-learning-based weather classification results and exploring climate scenarios. Combines VR, geospatial visualization, and machine learning to provide an interactive approach to understanding complex weather and climate data.

EDUCATION

- **Netaji Subhas University of Technology** *2020 – 2022*
  Master of Technology (M.Tech.) in Information Technology
  - Thesis Title: Design and Analysis of Certificateless Schemes for Healthcare Environment
  - Aggregate: 81.0% (GPA 8.1/10.0)
- **Govind Ballabh Pant University of Agriculture and Technology** *2017 – 2020*
  Bachelor of Technology (B.Tech.) in Computer Engineering
  - Project Title: Steg! - An Android Application for Metamorphic Cryptography
  - Aggregate: 72.4% (GPA: 7.2/10.0)

PERSONAL SKILLS

Technical Skills
- **Domain**: Queer GIS, Social XR and HCI, Digital Humanities and Critical Computing, Inclusive Technology, Machine Learning, Applied Cryptography
- **Languages**: C#, Python, C++, TypeScript
- **Tools**: Unity, React, Vite, Canvas, Latex, Github

Communication Skills	◦ Assisted in **conducting laboratory sessions** and tutorials, supporting students in understanding course concepts and practical implementations. ◦ **Guided students** with assignments, mini-projects, and problem-solving exercises. ◦ Helped in **preparing teaching materials** such as lecture slides, lab manuals, and tutorial sheets. ◦ Mentoring Interns in the Integrated Domain of Machine Learning and Virtual Reality.
Organizational Skills	◦ Served as an **organizer** for a f**ive-day technical workshop sponsored** by A**ll India Council for Technical Education**, coordinating planning, scheduling, and execution of technical sessions. ◦ Assisted in the **preparation of workshop materials**, registrations, and participant communications. ◦ Acted as an **organizing member** for the **inauguration event of a DPSU Bhawan**, supporting event planning and on-site coordination. ◦ **Coordinated ceremonial arrangements**, guest management, and technical arrangements for the inauguration event.
Certifications	◦ **Geographical Information System** by Department of Space, India ◦ **India in a Changing Climate** by Indian Institute of Technology Hyderabad, India. ◦ **Artificial Intelligence for Social Good** by Microsoft.

PUBLICATIONS

- Presented **5+ papers** in International and national conferences
- Published **2 papers** in SCI indexed Journals.
- Earlier publications are in applied cryptography and information security, which inform my current interest in privacy-preserving technologies for LGBTQ+ communities.
- GIS Based Real time Fire Detection and Anti Fire Simulation System.(Accepted, 47th Asian Conference on Remote Sensing (ACRS) 2026)

REFERENCES

1. **Shivani Arya**
   Senior Member Research Staff, CRL-BEL, Ministry of Defence, India
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   Relationship: Reporting Manager
2. **Dr. Devender Kumar Yadav**
   Associate Professor, NSUT, New Delhi, India
   Email: dk_iitm@yahoo.co.in
   Relationship: Master's Advisor